Introduction

This course is an introductory course in natural language processing, which has become highly interesting due to its everyday applications. It's important to consider the dual nature of NLP: some study it for its practical technology and implementation, while scientists focus on its scientific basis as computational models of language and modern intelligence.

Nowadays, products built on Natural Language Processing (NLP) have become integral parts of our daily lives, to the extent that it is hard to imagine living without them. Technologies like virtual assistants (e.g., Siri), simple workflows like auto-translating via machine translation frameworks, and newer technologies like writing assistants are ubiquitous.

In addition, entire sub-fields of NLP are integral to our day-to-day work. For example, information extraction aims to map text to a database or structured format. This process needs to be scalable, as we often parse large amounts of data. The accuracy of such extraction is domain-dependent, whether the domain is open or ontology-based, and the type of inference involved.

Text generation is another vital aspect of our lives. We use large language models (LLMs) for nearly everything, such as summarization tools, learning assistants, and generating course material for us. Of course, when discussing text generation, we cannot omit code generation, the tool that is increasingly replacing many of us. While it is now a field on its own, it initially emerged from natural language processing (NLP).

Natural Language Processing

We made clear that NLP is a vital part of our lives and thus has its right to exist. But what is NLP formally?

- Definition: Natural Language Processing (NLP) encompasses any task expressed in text, including verbal capabilities.
- Key Components:
  - Logical and "common sense" reasoning
  - Propositional knowledge
  - Conversational AI
- Note: This field is in a period of transition, so boundaries with related fields may shift.

Computers as Models for Linguistic Behavior

Beyond building practical consumer tools, NLP functions as a scientific framework to study the mechanisms of communication and cognition:

- Engineering vs. Human Intelligence:
  - System Design: Developing an intelligent artificial system requires computational models capable of handling complex linguistic layers—ranging from formal grammar and syntax to contextual semantics, pragmatic nuances, and world knowledge.
  - Comparative Constraints: Evaluating whether artificial systems and human learners face identical constraints. While modern artificial systems rely on massive parameters, extensive training datasets, and compute, human learners acquire language rapidly despite data scarcity (the poverty of the stimulus) while operating under biological memory limits and low energy footprints.
- Language as the Hallmark of Intelligence (The Turing Test):
  - Formulated by Alan Turing (1950), the Turing Test established natural language dialogue as the operational benchmark for machine intelligence.
  - Because fluid linguistic communication demands reasoning, context awareness, common-sense deduction, and adaptive conversation, mastering human language serves as a proxy for evaluating artificial cognitive capability.

Brief History of NLP

When looking at natural human language and computer language, it becomes apparent that they differ fundamentally: while human language is ambiguous, implicit, culturally loaded, and reliant on extensive world knowledge, computer language is strictly unambiguous, explicit, and grounded in formal logic.

Because of these stark contrasts, bridging the two is a non-trivial challenge. Looking at how this field evolved over time helps map out how researchers tackled this gap:

Brief History of NLP

- 1950s–1960s (Early Computing & Theory): Focused on theoretical work and proof-of-concept projects, pioneering machine translation experiments, and initial attempts to formalize linguistic theory.
- 1970s–1980s (Rule-Based Methods): Emphasized handcrafted rules and deeper models of syntax and semantics; confined largely to narrow "toy domains" that did not scale well.
- 1990s (Corpus-Based & Empirical Methods): Transitioned to statistical approaches using text corpora and empirical evaluation, often trading deep linguistic analysis for robust, real-world methods.
- 2000s (Statistical Machine Learning): Adopted sophisticated machine learning algorithms focused on scalability, data-driven pipelines, and practical applications.
- Present (Large Language Models): Dominated by LLMs setting the tone across the industry, with core focuses on multi modal capabilities, model scaling and efficiency, and advanced semantic tasks.

Speedy Progress in Recent Years

While modern breakthroughs—principally driven by machine learning—have fueled massive progress in long-standing challenges like machine translation and semantic disambiguation, natural language processing remains far from a solved discipline. Key open problems and active research frontiers persist across several fronts:

- Natural Language Interfaces for Knowledge Bases: Developing robust systems capable of directly querying, updating, and manipulating structured knowledge stores using conversational text.
- Mitigating Hallucination & Calibrating Uncertainty: Ensuring models report failure or state their uncertainty rather than generating plausible yet factually incorrect answers.
- Cross-Lingual Generalization: Building tools that reliably adapt to low-resource and typologically diverse global languages rather than remaining English-centric.
- Computational & Cost Efficiency: Designing lightweight, resource-effective models and inference techniques that deliver high utility without unsustainable compute, memory, or environmental costs.

This course bridges the gap between historical perspectives and contemporary innovations by examining both foundational classical methodologies and cutting-edge paradigms, maintaining a core emphasis on the theoretical and mathematical foundations of the field.

Summarization

A key frontier task in NLP is summarization: condensing text into a human-readable summary.

It splits into two main approaches:

- Extractive: Pulling important phrases directly from the source text as-is.
- Abstractive: Rewriting the text based on its core meaning.

Summarization is inherently difficult because it is heavily goal-dependent, making a "good" summary hard to define and evaluate. Even with recent progress, models still produce blatant errors like hallucinations.

Ambiguity and Redundancy

A major reason language is a hard computational problem is ambiguity. Language has rich underlying structure that is often ambiguously expressed across every level:

- Sounds (Phonology): Words can sound identical but mean completely different things (e.g., bare vs. bear).
- Morphemes (Morphology): The same word parts can convey multiple, distinct meanings.
- Sentences (Syntax & Semantics): Identical surface sentences can have wildly different interpretations.
  - Example: "The researcher watched a crocodile with a telescope" — Did the researcher use a telescope to observe, or did the crocodile have one?

Alongside ambiguity is the mirror issue of redundancy: there are countless different ways to express the exact same thought.

Headline Ambiguities in the Wild

Real-world headlines demonstrate how easily syntax and word choice create unintended interpretations:

- "Teacher strikes idle kids"
- (Are teachers hitting unsupervised children, or are labor strikes causing kids to stay home?)
- "Hospitals are sued by 7 foot doctors"

- (Seven podiatrists suing, or doctors who are literally seven feet tall?)
- "Iraqi head seeks arms"
- (A government official requesting weapons, or a decapitated head looking for body parts?)
- "Stolen painting found by tree"
- (Discovered near a tree, or did a tree track down the stolen art?)
- "Kids make nutritious snacks"
- (Children preparing healthy food, or children being consumed as healthy snacks?)
- "Local HS dropouts cut in half"
- (Dropout rates reduced by 50%, or students literally sawed in half?)

Structural Ambiguity & The Limits of Data

Ambiguity is far worse than it appears because human brains subconsciously discard structurally valid but nonsensical parses:

- Example:[Rock and Roll] concerts (concerts of the musical genre)
  - Rock and [Roll concerts] (literal stones, plus concerts that physically roll)

Can't we resolve this by simply observing enough examples of "rock and roll"? Not easily, because of two core obstacles: variation (redundancy) and data sparsity.

Variation & Paraphrasing: One Meaning, Many Forms

There are endless ways to express the exact same concept:

- Monty Python's Dead Parrot sketch: "He's passed on", "This parrot is no more", "He has ceased to be", "He's expired", "Pushed up the daisies", "Kicked the bucket", "Joined the choir invisible" → all map to the single state: dead.
- Questions: "What does X do for a living?", "What is X's profession?", "What does X do?" → all ask for the same attribute: occupation.

Recognizing that different surface expressions share identical underlying semantics is critical for tasks like Information Extraction (IE) and Question Answering (QA).

Data Sparsity & Zipf's Law

Data Sparsity means that even in massive text corpora, most possible combinations of words are seen rarely or never at all. You can never observe "enough examples" to cover all valid language use.

This is governed by Zipf's Law:

$$\log f + \log r = k \quad (\text{or } f \cdot r = C)$$

- f (frequency): How many times a word occurs in a corpus.
- r (rank): The word's position when all words are ordered from most to least frequent (r=1 for the most frequent word, e.g., "the").
- k (constant): A constant specific to the corpus.

Key takeaway: A tiny handful of words make up most of what we read, while the vast majority of words sit in a massive "long tail" occurring only once or twice. Sparsity is a permanent condition in NLP, affecting vocabulary, syntactic parses, and semantic patterns alike.

Many Rules, Many Exceptions

Hard-coded linguistic rules fail because natural language is riddled with arbitrary exceptions and historical quirks:

- Morphology: Regular past tense adds -ed (walk → walked), but irregulars break the rule entirely (go → went, sing → sang).
- Collocations & Idioms: "I haven't a clue" is natural English, but "I haven't a look" is completely ungrammatical.
- Tense vs. Time: "The tour begins at 14:40 today" uses present tense for a future scheduled event, but "He is not home at 17:00" indicates a habitual routine; a prediction requires "He will not be home at 17:00".
- Syntax/Word Order: We say "River Thames" (noun after) but "Mississippi River" (noun before).

The Foundational Debate: Rules vs. Statistics

Now that we understand the field and what makes its problems hard, consider these two opposing perspectives on how language works:

- Noam Chomsky (1969) — The Rule-Based View:
- "It must be recognized that the notion 'probability of a sentence' is an entirely useless one, under any known interpretation of this term."
  - Argues that language is governed by formal syntactic rules, not statistical likelihood; a sentence can be completely novel and unheard of, yet entirely grammatical.
- Claude Shannon (1951) — The Statistical View:
- "...anyone speaking a language possesses, implicitly, an enormous knowledge of the statistics of the language. Familiarity with the words, idioms, clichés and grammar enables him to fill in missing or incorrect letters in proof-reading, or to complete an unfinished phrase in conversation."
  - Argues that language comprehension relies on implicit statistical knowledge, allowing speakers to predict words, correct errors, and resolve ambiguity based on probability.

Layers of Linguistic Representation

Language consists of multiple structural layers that humans effortlessly integrate during production and comprehension. Using the example "This is a simple sentence" (and contrasting it with "But it is an instructive one"), we can break down these core representational levels:

- Surface Text (Tokens): The raw string of words: This, is, a, simple, sentence.
- Parts of Speech (POS Tags): The syntactic category of each word:
  - det (determiner / DT), verb (VBZ), det (DT), adj (adjective / JJ), noun (NN).
- Morphology: The grammatical features of individual words, such as lemma, person, number, and tense (e.g., is → root be, 3rd person singular, present tense).
- Syntax (Constituency Parse Tree): How words group into hierarchical phrases to form a valid sentence (S):
  - S → NP + VP
  - NP → DT (This)
  - VP → VBZ (is) + NP (a simple sentence)
  - NP → DT (a) + JJ (simple) + NN (sentence)
- Semantics (Word Meaning): The definitions and concepts behind individual words:
  - simple → "having few parts"
  - sentence → "string of words satisfying the grammar of a language"
- Discourse / Pragmatics: Relationships beyond the single sentence (e.g., establishing a contrast across clauses: "This is a simple sentence" $\xrightarrow{\text{Contrast}}$ "But it is an instructive one").

The course is structured around core NLP application tasks paired with the computational and statistical methods used to solve them:

Natural Language Processing Tasks	**Foundational & Modern Methods**
• Language Models	• Probabilistic Modeling (Generative vs. Discriminative models)
• Document Classification	• Naïve Bayes Classifiers
• Part-of-Speech (POS) Tagging	• Log-Linear Classifiers
• Syntax & Parsing (graph-based methods)	• Feed-Forward Neural Networks
• Sentiment Analysis	• Markov Models & Hidden Markov Models (HMMs)
• Semantic Analysis	• Conditional Random Fields (CRFs)
• Information Extraction	• Structured Prediction
• Machine Translation	• Recurrent Neural Networks (RNNs)
	• Transformers & Pretraining

Useful Resources

Textbooks & Course Notes

- Speech and Language Processing (Dan Jurafsky & James H. Martin): The standard reference textbook for NLP and speech recognition (3rd edition drafts freely accessible online).
- Introduction to Natural Language Processing (Jacob Eisenstein): A rigorous textbook covering classical statistical formulations through neural architectures (free open-access draft available on GitHub).
- HUJI NLP 67658 Course Materials (aliKashoa-cs): Course notes, assignments, and local reference repository

for Hebrew University's NLP (67658).

Literature & Research Repositories

- ACL Anthology: The primary open-access digital archive for computational linguistics and NLP research papers.
  - Top Conferences: ACL, EMNLP, NAACL, EACL, and CoNLL.
  - Top Journals: TACL (Transactions of the Association for Computational Linguistics) and Computational Linguistics.
- Machine Learning Conferences: High-impact NLP breakthroughs are regularly presented at general ML venues, including NeurIPS, ICML, and ICLR.
- Preprints & Search Engines: Use arXiv (cs.CL) for cutting-edge preprints, alongside Google Scholar and Semantic Scholar for literature tracking and citation graphs.